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PROJECT TASK - 3

DOCUMENTATION

REPORT

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D Flip-Flop & 4-Bit Counter Build

Abstract :

This project implements a D Flip-Flop and a 4-Bit Binary Counter using Verilog HDL. The D Flip-Flop is a sequential logic circuit used for storing 1-bit data based on clock pulses. The binary counter increments its count value during every positive edge of the clock signal.

The project demonstrates the working of sequential digital circuits, clock synchronization, and binary counting operations. Simulation and waveform verification were performed using EDA Playground and EPWave.

Objective :

The main objective of this project is:

- To design a D Flip-Flop using Verilog HDL
- To implement a 4-Bit Binary Counter
- To understand sequential circuit operation
- To perform simulation and waveform analysis
- To verify counter increment functionality using clock pulses

Tools Used

Tool	Purpose
Verilog HDL	Hardware Discription Language
EDA Playground	Online Simulation Platform
Icarus Verilog	Verilog Simulator
EPWave	Waveform Viewer

Theory

D Flip-Flop :

A D Flip-Flop is a memory element used to store one bit of data. The output changes only at the positive edge of the clock signal.

Operation :

- If clock edge occurs:

$$Q = D$$

- The output retains its previous state until the next clock pulse.

4-Bit Binary Counter :

A binary counter increments its output value by 1 on every positive clock edge.

Counting Sequence :

0000

0001

0010

0011

0100

0101

The counter continuously performs binary counting operations.

Working Principle

- The clock signal toggles continuously.
- The D Flip-Flop captures input data during the positive edge of the clock.
- The binary counter increments automatically during each positive clock pulse.
- Simulation waveforms verify correct circuit functionality.

Verilog Modules Used

D Flip-Flop Module :

- Stores 1-bit data
- Triggered by positive clock edge

Counter Module :

- 4-bit sequential counter
- Increments value automatically

Flies Included

- 📌 Main Verilog design code
- 📌 Testbench code
- 📌 Output screenshots
- 📌 EPWave waveform screenshots

Simulation Procedure

- Open EDA Playground
- Select Verilog language
- Paste design code into design.sv
- Paste testbench into testbench.sv
- Click Run
- Open EPWave
- Add signals:
 1. Clk
 2. d
 3. q
 4. Count



Output Waveform Analysis

The waveform output shows:

- Clock pulse generation
- Data input changes
- Flip-Flop output updates
- Counter increment operation

The counter increments correctly at every positive clock edge.

Example :

Clock Edge	Counter Output
1	0001
2	0010
3	0011
4	0100

TestBench Code,Main Code

The screenshot displays the EDA Playground interface with two code editors. The left editor, 'testbench.sv', contains a testbench for a 4-bit counter. The right editor, 'design.sv', contains the main Verilog code for a D Flip-Flop and a 4-bit counter module. The bottom panel shows the simulation log with a sequence of clock edges and the resulting counter output.

testbench.sv

```
1 // Testbench for Flip-Flop and Counter
2 module testbench;
3
4 // Inputs
5 reg clk;
6 reg d;
7
8 // Outputs
9 wire q;
10 wire [3:0] count;
11
12 // Instantiate D Flip-Flop
13 d_flipflop uut1(
14     .clk(clk),
15     .d(d),
16     .q(q)
17 );
18
19 // Instantiate Counter
20 counter uut2(
21     .clk(clk),
22     .count(count)
23 );
```

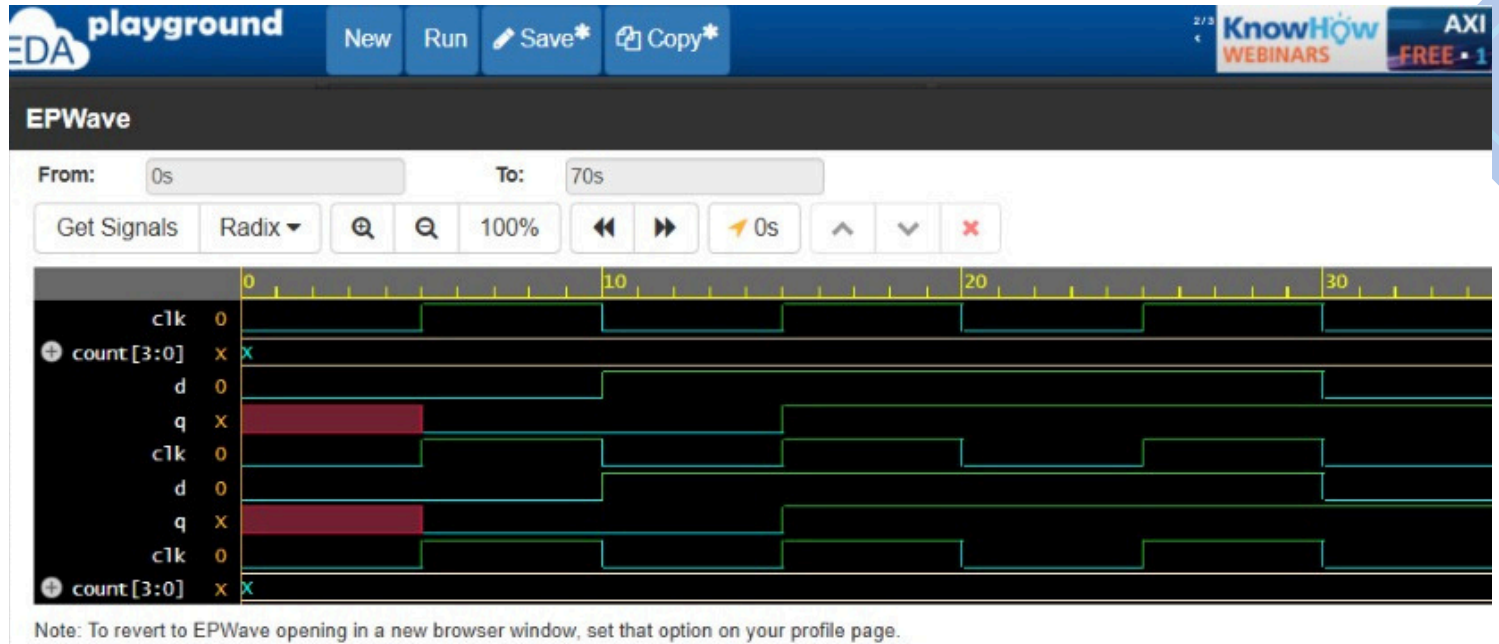
design.sv

```
1 // D Flip-Flop Module
2 module d_flipflop(
3     input clk,
4     input d,
5     output reg q
6 );
7
8 // Trigger on positive clock edge
9 always @(posedge clk)
10 begin
11     q <= d;
12 end
13 endmodule
14
15 // 4-Bit Counter Module
16 module counter(
17     input clk,
18     output reg [3:0] count
19 );
```

Simulation Log

```
Time=40 clk=0 d=0 q=0 count=xxxx
Time=45 clk=1 d=0 q=0 count=xxxx
Time=50 clk=0 d=1 q=0 count=xxxx
Time=55 clk=1 d=1 q=1 count=xxxx
Time=60 clk=0 d=1 q=1 count=xxxx
Time=65 clk=1 d=1 q=1 count=xxxx
testbench.sv:65: $finish called at 70 (1s)
Time=70 clk=0 d=1 q=1 count=xxxx
Finding VCD file...
./dump.vcd
[2026-06-04 09:02:03 UTC] Opening EPWave...
Done
```

Logic Gates Design-EV Waveforms



References :

- Verilog HDL Documentation
- Digital Electronics Fundamentals
- EDA Playground Simulator Documentation

Conclusion :

The D Flip-Flop and 4-Bit Binary Counter were successfully designed and simulated using Verilog HDL. The waveform outputs verified correct sequential circuit behavior and counter operation. The project helped in understanding clock-based digital systems and Verilog HDL implementation techniques.